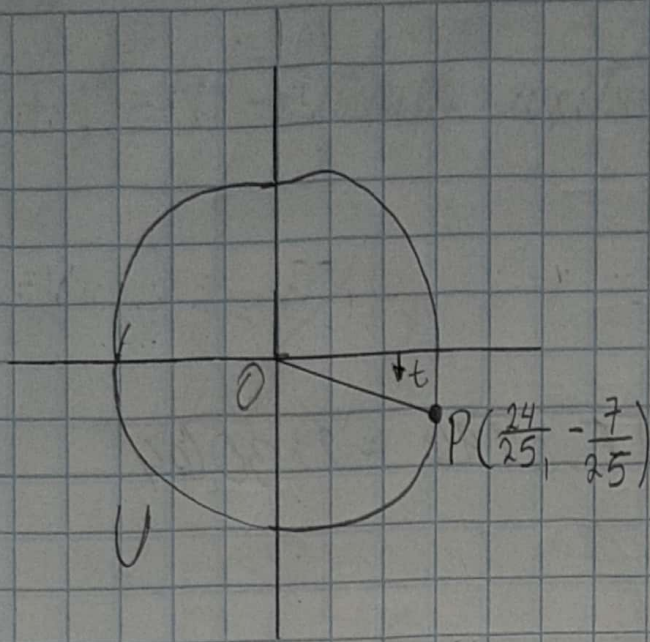


Tarea Sección 5.3

3)



$$\sec t = \frac{-7}{25} \quad \csc t = \frac{25}{-7}$$

$$\cos t = \frac{24}{25} \quad \sec t = \frac{25}{24}$$

$$\tan t = \frac{-7/25}{24/25} = \frac{-7}{24}$$

$$\cot t = \frac{24}{-7}$$

7) $(-\frac{12}{13}, -\frac{5}{13})$

a) $P(t + \pi)$

$$\left(\frac{12}{13}, \frac{5}{13}\right)$$

b) $P(t - \pi)$

$$\left(\frac{12}{13}, \frac{5}{13}\right)$$

c) $P(-t)$

$$\left(-\frac{12}{13}, \frac{5}{13}\right)$$

d) $P(-t - \pi)$

$$\left(\frac{12}{13}, -\frac{5}{13}\right)$$

15) a) $5\pi/4$

$$\begin{aligned} x^2 + y^2 &= 1 \\ 2x^2 &= 1 \\ x^2 &= \frac{1}{2} \\ x &= \frac{\sqrt{2}}{2} \end{aligned}$$

$$P = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$$

$$\sin 5\pi/4 = -\frac{\sqrt{2}}{2}$$

$$\csc 5\pi/4 = \frac{\sqrt{2}}{2}$$

$$\cos 5\pi/4 = -\frac{\sqrt{2}}{2}$$

$$\sec 5\pi/4 = -\frac{\sqrt{2}}{2}$$

$$\tan 5\pi/4 = 1$$

$$\cot 5\pi/4 = 1$$

19) a) $\cot(-\frac{3\pi}{4}) = 1$

b) $\sec(-45^\circ) = \sqrt{2}$

c) $\csc(-\frac{3\pi}{2}) = 1$

25) $\frac{1}{\cos(-x)} - \tan(-x) \sin(-x) = \cos x$

$$\frac{1}{\cos(x)} - (-\tan(x))(-\sin(x))$$

$$\frac{1}{\cos(x)} - \frac{\sin(x)}{\cos(x)} (\sin(x))$$

$$\frac{1}{\cos(x)} - \frac{\sin^2(x)}{\cos(x)}$$

$$\frac{1 - \sin^2(x)}{\cos(x)} = \frac{\cos^2(x)}{\cos(x)} = \cos(x)$$

35) a) Como $x \rightarrow (\pi/2)^+$, $\sec x \rightarrow \infty$

b) Como $x \rightarrow 0$, $\sec x \rightarrow 1$

37) a) Como $x \rightarrow 0^-$, $\csc x \rightarrow -\infty$

b) Como $x \rightarrow \pi/4$, $\csc x \rightarrow \sqrt{2}$

45) $\cos x = \sqrt{2}/2$ $[0, 4\pi]$

$\frac{\pi}{4}$

$\frac{7\pi}{4}$

$\frac{9\pi}{4}$

$\frac{15\pi}{4}$

49) $\tan x = 0$ $(-\pi/2, 3\pi/2)$

0

π

53) $y = \cos x; [-2\pi, 2\pi]; a = -\frac{1}{2}$

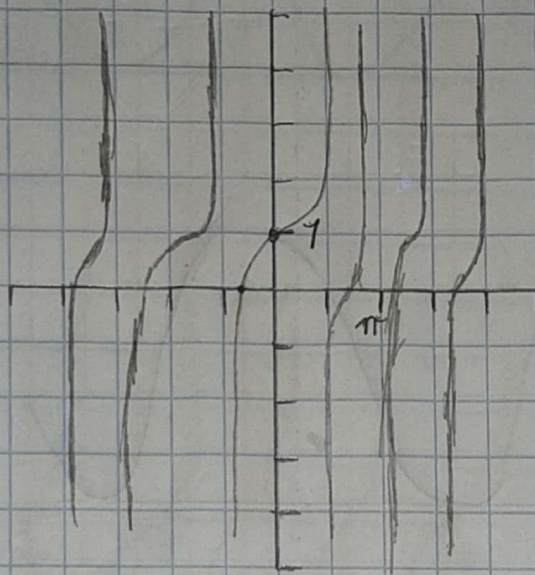
a) $\underline{-\frac{4\pi}{3}, -\frac{2\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}}$

b) $-2\pi \leq x < -\frac{4\pi}{3}, -\frac{2\pi}{3} < x < \frac{2\pi}{3}, \frac{4\pi}{3} < x \leq 2\pi$

c) $\underline{-\frac{4\pi}{3} < x < -\frac{2\pi}{3}, \frac{2\pi}{3} < x < \frac{4\pi}{3}}$

59) $y = 1 + \tan x$

$P(0, 1)$



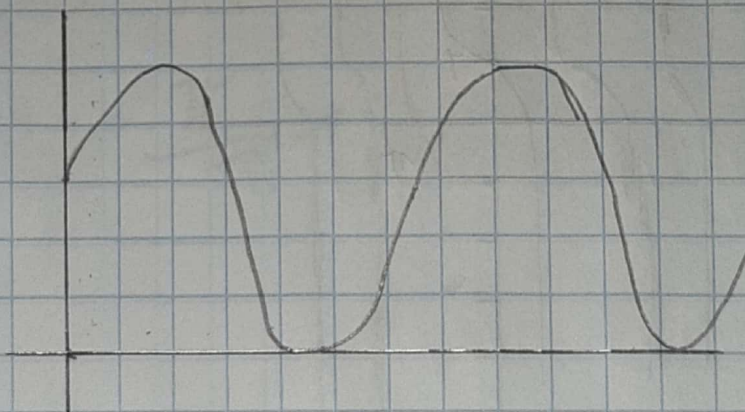
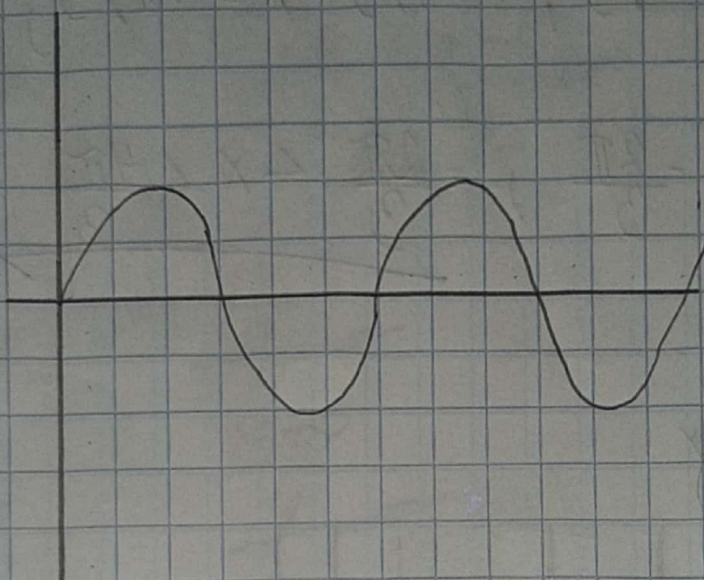
65) tangente intervalo -2π y 2π

a) aumenta $= [-2\pi, -\frac{3\pi}{2}), (-\frac{3\pi}{2}, -\frac{\pi}{2}), (-\frac{\pi}{2}, \frac{\pi}{2}), (\frac{\pi}{2}, \frac{3\pi}{2}), (\frac{3\pi}{2}, 2\pi]$

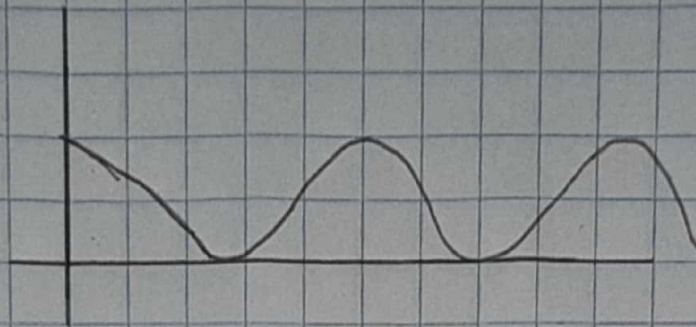
todos los valores donde no se define

b) disminuye) = tangente incrementa en todos los intervalos
siempre que no se indefina, más no disminuye.

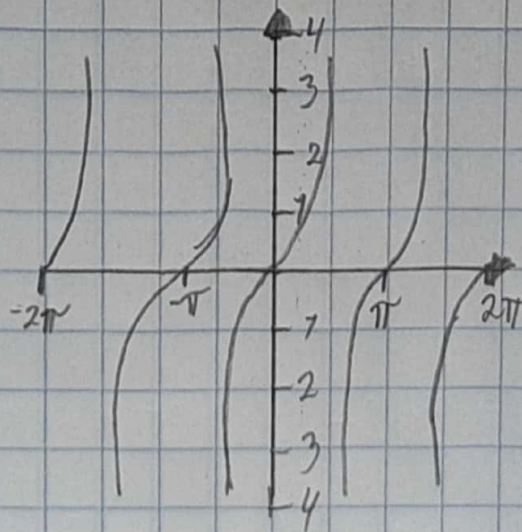
67) seno



coseno



tangente



71) a) $\cos 4 = \underline{-0.7}$

b) $\cos(-1.2) = \underline{0.4}$

c) Todos los números t entre 0 y 2π , de modo que $\cos t = -0.6 = \underline{2.2, 4.7}$